Using AI to improve your code

Eris :3 (they/them)

October 20, 2026

Using ~~AI~~ Abstract Interpretation to improve your code

Eris :3 (they/them)

October 20, 2026

How other people use Abstract Interpretation to improve your code when you run it

Eris :3 (they/them)

October 20, 2026

I FUCKING HATE AI!!!

I FUCKING HATE AI!!!

Please imagine a video of Sam Altman being blown up by a missile.

I FUCKING HATE AI!!!

Please imagine a video of Sam Altman being blown up by a missile.
These are my thoughts on ai :)

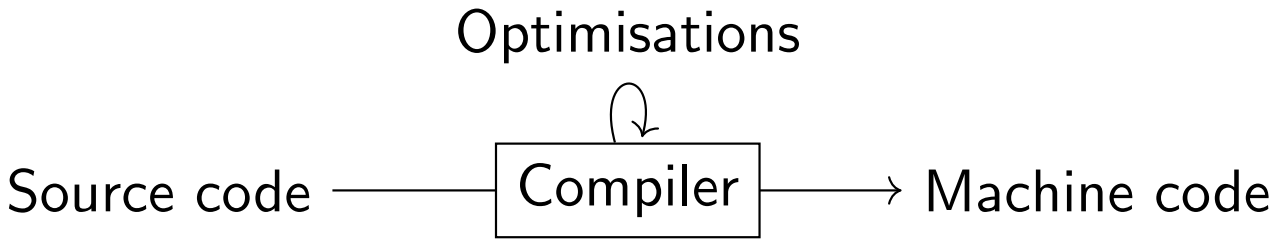
I FUCKING HATE AI!!!

Please imagine a video of Sam Altman being blown up by a missile.
These are my thoughts on ai :) Artificial intelligence will not be mentioned again for the rest of this talk.

Optimising compilers



Optimising compilers



Suboptimal program

$F \leftarrow \underbrace{[0, \ldots, 0]}_{\text{100 of them}}$
$F[0] \leftarrow 0$
$F[1] \leftarrow 1$
for $i \leftarrow 2, \ldots, 99$ **do**
    $F[i] \leftarrow F[i-1] + F[i-2]$
end for

Suboptimal program

$F \leftarrow \underbrace{[0, \ldots, 0]}_{\text{100 of them}}$
if $\neg(0 \leq 0 \leq 99 \wedge 0 \leq 1)$ **then**
    explode (out of bounds)
end if
$F[0] \leftarrow 0$
$F[1] \leftarrow 1$
for $i \leftarrow 2, \ldots, 99$ **do**
    if $\neg(0 \leq i - 2 \leq 99 \wedge 0 \leq i - 1 \leq 99 \wedge 0 \leq i \leq 99)$ **then**
        explode (out of bounds)
    end if
    $F[i] \leftarrow F[i - 1] + F[i - 2]$
end for

Suboptimal program

$F \leftarrow \underbrace{[0, \ldots, 0]}_{\text{100 of them}}$
if $\neg(0 \leq 0 \leq 99 \wedge 0 \leq 1)$ **then**
    explode (out of bounds)
end if
$F[0] \leftarrow 0$
$F[1] \leftarrow 1$
for $i \leftarrow 2, \ldots, 99$ **do**
    if $\neg(0 \leq i-2 \leq 99 \wedge 0 \leq i-1 \leq 99 \wedge 0 \leq i \leq 99)$ **then**
        explode (out of bounds)
    end if
    $F[i] \leftarrow F[i-1] + F[i-2]$
end for

The bounds check is redundant! How could we automatically determine this though?